

Sean M. Keogh
National Museum of Natural History
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EDUCATION

- 2023 **Ph.D.**, University of Minnesota
Ecology, Evolution, and Behavior
Dissertation: Natural selection & phenotypic plasticity shape freshwater mussel ecomorphological evolution
- 2018 **M.S.**, University of Minnesota
Conservation Sciences
Thesis: Integrative species delimitation of *Lampsilis teres* species complex (Bivalvia: Unionidae)
- 2014 **B.S.**, University of Minnesota
Fisheries, Wildlife, and Conservation Biology
Undergrad research: Population dynamics of imperiled freshwater mussels & their host fishes

APPOINTMENTS

- 2025-Present Peter Buck Postdoctoral Fellow
Smithsonian Institution – National Museum of Natural History
- 2023-Present Museum Associate
Bell Museum of Natural History – University of Minnesota
- 2025-Present Collection Associate
Field Museum of Natural History
- 2023-2025 Collections Manager of Invertebrate Zoology
Field Museum of Natural History

FUNDING

Research Fellowships

- 2025 National Museum of Natural History – Peter Buck Fellowship. \$144,000
- 2022 Bell Museum of Natural History – Simons Fellowship in Systematic Biology. \$21,637
- 2021 National Museum of Natural History – Smithsonian Institution Predoctoral Fellowship. \$40,000
- 2020 UMN – Interdisciplinary Doctoral Fellowship. \$25,000
- 2020 UMN – Ecology, Evolution, and Behavior Summer Fellowship. \$6,461
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- 2019 Bell Museum of Natural History – Dayton Fellowship. \$25,000
- 2017 UMN – Moos Graduate Research Fellowship in Aquatic Biology. \$7,340
- 2016 Bell Museum of Natural History – Joyce Davenport Fellowship. \$5,917
- 2015 UMN – John Dobie Fellowship. \$16,218

Research Grants

- 2025 FMNH – Grainger Bioinformatics Center Research Award. \$8,500
- 2023 Linnean Society of London – LinnéSys: Systematics Research Fund. \$1,255
- 2021 National Museum of Natural History – Smithsonian Institution Predoc Research Grant. \$4,000
- 2021 Bell Museum of Natural History – Dayton Grant. \$2,000
- 2021 Society of Integrative & Comparative Biology – Grants-In-Aid of Research. \$1,000

- 2021 UMN – Ecology, Evolution, and Behavior Summer Award. \$1,500
 2020 UMN – Ecology, Evolution, and Behavior Summer Award. \$1,500
 2020 UMN – Florence Rothman Research Grant. \$2,000
 2020 The Society for Systematic Biologists Graduate Research Award. \$2,000
 2020 Bell Museum of Natural History – Davenport Grant. \$2,500
 2019 Bell Museum of Natural History – Davenport Grant. \$2,500
 2019 UMN – Ecology, Evolution, and Behavior Summer Award. \$1,500
 2018 Minnesota Department of Natural Resources Research Contract 146124/3000139381. \$2,505
 2018 American Malacological Society – Melbourne R. Carriker Student Award. \$1,000
 2018 Conchologists of America Academic Grant. \$2,300
 2017 UMN – Wally Dayton Wildlife Fund. \$2,500
 2017 UMN – Moos Graduate Research Grant. \$2,000
 2016 Bell Museum of Natural History Dayton Grant. \$1,500
 2013 UMN – Undergraduate Research Opportunities Program Grant. \$1,700

Travel Grants

- 2024 FMNH – Negaunee Field Work Fund. \$5,000
 2022 UMN – Ecology, Evolution, and Behavior Travel Award. \$900
 2019 UMN – Ecology, Evolution, and Behavior Travel Award. \$900
 2018 UMN – Conservation Sciences Travel Grant. \$350
 2017 UMN – Conservation Sciences Travel Grant. \$350

PUBLICATIONS (*student coauthor)

Manuscripts in submission, review, or revision

Keogh, S., *Chmielewski, E., Minerich, B., & Stapleton, S. Mesocosm experiments shed light on the ecological drivers of shell plasticity in a cohort of sibling freshwater mussels. *In review*.

Haugh, J.* †, **Keogh, S.**†, Pfeiffer, J., & Zanatta, D. Host infection explains sexual dimorphism in a freshwater mussel clade on the edge of extinction (Unionidae: *Epioblasma*). *In review*. †Authors share first authorship.

Pfeiffer, J., Hampson, E., Guerra Canedo, V., Goetz, F., **Keogh, S.**, Froufe, E., Lopes-Lima, M., Gome-dos-Santos, A., Johnson, N., Smith, C., Bogan, A., Klunzinger, M., Jones, H., Ziertz, A. The evolution of cementation in freshwater mussels (Bivalvia: Unionoida). *In revision for resubmission*.

Published

11. Lopez, J., Hopper, G., Sánchez González, I., **Keogh, S.**, Bucholz, J., Lozier, J., & Atkinson, C. Leveraging the elementome to test niche-based and stoichiometric hypotheses. Ecological Monographs, *Accepted*.
10. **Keogh, S.**, *Celli, S., & *Ramos, A. 2026. Ecological interactions between imperilled molluscs: *Somatogyrus* pebblesnail oviposition on freshwater mussels. Journal of Molluscan Studies, 92, eyag021. <http://doi.org/10.1093/mollus/eyag021>
9. Hopper, G., **Keogh, S.**, Atkinson, C., DuBose, T., Lozier, J., Pfeiffer, J., & Sánchez González, I. 2026. Linking body size, dispersal, range size, and fragmentation to assess extinction risk of freshwater mussels. Journal of Biogeography, 53, e70277. <http://doi.org/10.1111/jbi.70277>
8. Franzen, A., Sansom, B., Pfeiffer, J., Goetz, F., **Keogh, S.**, & Vaughn, C. 2026. Evaluating correlations between freshwater mussel shell morphology and valve-gaping activity across riverine conditions. Freshwater Biology, 71, e70257. <http://doi.org/10.1111/fwb.70257>

7. **Keogh, S.**, Johnson, N., Smith, C., Sietman, B., Garner, J., Randklev, C., & Simons, A. 2025. Secondary contact erodes Pleistocene diversification in a wide-ranging freshwater mussel (*Quadrula*). Molecular Ecology, 34, e17572. <http://doi.org/10.1111/mec.17572>
6. **Keogh, S.**, Minerich, B., Ohlman, L., Pletta, M., Scheunemann, A., Schroeder, Z., Secrist, Z., Franzen, A., Sietman, B., & Simons, A. 2025. Phenotypic plasticity, multiple paternity, and shell shape divergence across lake-stream habitats in a freshwater mussel brood (*Pyganodon grandis*). Freshwater Biology, 70, e14358. <http://doi.org/10.1111/fwb.14358>
5. Pfeiffer, J., DuBose, T., & **Keogh, S.** 2024. Synthesis of natural history collections data reveals patterns of US freshwater mussel diversity and decline. Biological Conservation, 291, 110462. <https://doi.org/10.1016/j.biocon.2024.110462> [Press](#) [MusselMapR](#)
4. **Keogh, S.**, Pfeiffer, J., Simons, A. & Edie, S. 2024. Riverine flow rate drives widespread convergence in the shell morphology of imperiled freshwater mussels. Evolution, 78, 39-52. <https://doi.org/10.1093/evolut/qpad190>
3. Hopper, G., Bucholz, J., DuBose, T., Fogelman, K., **Keogh, S.**, Kubala, M., Lodato, M., Nichols, D., Sánchez González, I., Pfeiffer, J., Stoeckel, J., Lozier, J., & Atkinson, C. 2023. A trait dataset for freshwater mussels of the United States of America. Scientific Data, 10, 745. <https://doi.org/10.1038/s41597-023-02635-9>
2. **Keogh, S.**, Johnson, N., Williams, J., Randklev, C., & Simons, A. 2021. Gulf Coast vicariance shapes phylogeographic history of a North American freshwater mussel species complex (Bivalvia: Unionidae). Journal of Biogeography, 48, 1138-1152. <https://doi.org/10.1111/jbi.14066>
1. **Keogh, S.** & Simons, A. 2019. Molecules and morphology reveal ‘new’ widespread North American freshwater mussel species (Bivalvia: Unionidae). Molecular Phylogenetics and Evolution, 138, 182-192. <https://doi.org/10.1016/j.ympev.2019.05.029>

NON PEER-REVIEWED ARTICLES & SPECIES ASSESSMENTS

7. **Keogh, S.** 2026. *Nephronaias fluckei*. The IUCN Red List of Threatened Species. *In review*.
6. **Keogh, S.** 2026. *Nephronaias gundlachi*. The IUCN Red List of Threatened Species. *In review*.
5. **Keogh, S.** 2026. *Nephronaias rowellii*. The IUCN Red List of Threatened Species. *In review*.
4. White, D., Badgett, N., **Keogh, S.**, Stegmann, E., & Dunn, H. 2021. DNA confirmation of a reproducing population of federally endangered Winged Mapleleaf (*Quadrula fragosa*) in the Little River, Oklahoma. Ellipsaria, 23(2), 27-30.
3. Scheunemann, A., Sietman, B., Hove, M., **Keogh, S.**, Pletta, M., Ohlman, L., Carrow, K., Rounds, C., & Taylor, S. 2020. Suitable Host Fishes for the Canary Kingshell, *Lampsilis sietmani* Keogh and Simons 2019. Ellipsaria, 22(1), 26-28.
2. **Keogh, S.** & Simons, A. 2019. Species and distributional boundaries of *Tritogonia nobilis* (Conrad, 1854) (Bivalvia: Unionidae). Minnesota Department of Natural Resources Report. Executed contract 146124/3000139381.
1. Hove, M., Sietman, B., Berg, M., Bump, S., Davis, M., Hansen, H., **Keogh, S.**, Luebke, C., Marr, S., Maynard, A., Murphy, K., Secrist Z., and Hornbach, D. 2013. *Pleurobema sintoxia* Early Life History. Ellipsaria, 15(3), 14-16.

PRESENTATIONS

Invited Seminars

- 2026 Howard University – Department of Biology
- 2024 Minnesota Zoo – Department of Conservation and Research
- 2023 University of Illinois – Department of Natural Resources and Environmental Sciences
- 2023 Chicago Shell Club
- 2023 Smithsonian National Museum of Natural History – Department of Invertebrate Zoology (*Virtual*)
- 2022 Field Museum of Natural History (*Virtual*)
- 2021 Harvard University – Museum of Comparative Zoology (*Virtual*)
- 2021 Regis University – Department of Biology
- 2019 Smithsonian National Museum of Natural History – Department of Invertebrate Zoology
- 2019 Auburn University Museum of Natural History
- 2018 University of Minnesota – Conservation Sciences

Conference Presentations

- 2026 Chesapeake Bay Foundation Freshwater Mussel Science Symposium. Charles City, VA. (*Poster*)
- 2025 Freshwater Mollusk Conservation Society. Ann Arbor, MI.
- 2023 Freshwater Mollusk Conservation Society. Portland, OR.
- 2022 Society for Integrative & Comparative Biology. Phoenix, AZ. (*Virtual*)
- 2020 Society for Integrative & Comparative Biology. Austin, TX. (*Poster*)
- 2019 Freshwater Mollusk Conservation Society. San Antonio, TX. (*Oral & Poster*)
- 2018 American Malacological Society. Oahu, HI. **Runner-up: Best Student Presentation Award**
- 2017 Freshwater Mollusk Conservation Society. Cleveland, OH. (*Poster*)

Co-authored Student Presentations

- 2026 C. Dekker. UNM McNair Scholars Research Conference. Albuquerque, NM
Poster – Don't move a mussel: Tracking the invasion history of a freshwater mussel using DNA barcoding
- 2026 K. Ghimire. Society for the Study of Evolution. Cleveland, OH
Poster – Shell to sequence: A genomic investigation into the native North American mussel, Fusconaia flava

TEACHING

Guest Instructor

- American University – Research Methods in Biological Sciences – Spring 2026
- Shedd Aquarium – Freshwater Ecology – Fall 2023

Laboratory Instructor (Teaching Assistant)

- UMN – Ornithology – Spring 2020 (2 sections)
- UMN – Ichthyology – Fall 2018 (2 sections)
- UMN – Ichthyology – Fall 2017 (2 sections)
- UMN – Ichthyology – Fall 2016 (2 sections)

MENTORSHIP

Graduate Students

- 2025- K. Ghimire – Loyola University Chicago (MS committee)
- 2025 C. Mejía-Rivera – FMNH Visiting Scholar (FMNH sponsor; \$3900 awarded to CMR)

Undergraduate Students

- 2026 C. Dekker – Cal Poly Humboldt & McNair Scholar (NHRE Intern)
- 2025 S. Celli – Miami University (FMNH Women in Science Intern)
- 2025 A. Ramos – Middlebury College (FMNH Women in Science Intern)

2024 K. Wilson – Chicago State University (ILSAMP Intern)
 2019 S. Sackner-Bernstein – Macalester College
 2019 M. Nord – Macalester College
 2018 A. Franzen – University of Minnesota
 2018 A. Rogowski – Macalester College
 2017 M. Buse – Macalester College
 2017 M. Glasenapp – Macalester College
 2017 J. Fernandez – Macalester College

High School Students

2025 L. Gallaga – FMNH Behind-the-Scenes Intern
 2025 M. De Alba – FMNH Behind-the-Scenes Intern
 2024 S. Morales – FMNH Behind-the-Scenes Intern
 2024 M. Hanley – FMNH Behind-the-Scenes Intern
 2024 M. Jung – FMNH Behind-the-Scenes Intern
 2021 F. Mohamed – Bell Museum HS Ecology Career Exploration Intern
 2021 K. Her – Bell Museum HS Ecology Career Exploration Intern
 2021 I. Rosenberg-Scholl – Bell Museum HS Ecology Career Exploration Intern
 2021 S. Pesola-Buck – Bell Museum HS Ecology Career Exploration Intern
 2021 Y. Wehelie – Bell Museum HS Ecology Career Exploration Intern
 2021 C. Yang – Bell Museum HS Ecology Career Exploration Intern

Volunteers

2025 C. Stein – FMNH Invertebrate Zoology Volunteer (Digitization)
 2025 L. Lewis – FMNH Invertebrate Zoology Volunteer (Digitization)
 2024-25 A. Sánchez – FMNH Invertebrate Zoology Volunteer (Digitization)
 2023-25 G. Gagne – FMNH Invertebrate Zoology Volunteer (Imaging)
 2023-25 T. Durica – FMNH Invertebrate Zoology Volunteer (Curation)
 2023-25 C. Bonbrest – FMNH Invertebrate Zoology Volunteer (Curation)

Non-academic mentorship

2026- Big Brothers Big Sisters mentor

COMMUNITY ENGAGEMENT

Citizen science

2025- [Snail Scribe Archive](#) – Zooniverse: Digitizing FMNH Invert Zoology Collection records

Outreach events

2026 National Museum of Natural History – Ocean Hall-O-Ween
 2026 Georgetown University Department of Biology – Career spotlight lunch & learn
 2025 Educator Open House at Field Museum – Freshwater mussels: the lives of our rivers
 2025 Meet a Scientist at Field Museum Science Hub – Freshwater mussels: the lives of our rivers
 2025 Field Museum Members' Night – Discover the vast diversity of invertebrate zoology
 2024 Field Museum Learning Center – Incorporating snail shells into K-12 education
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 2019 Conservation Corps Youth Outdoors Career Day – Bell collection tours & career discussion
 2018 Conservation Corps Youth Outdoors Career Day – Bell collection tours & career discussion
 2018 Science Sunday – Public engagement at Bell Museum mollusk exhibit
 2017 Conservation Corps Youth Outdoors Career Day – Bell collection tours & career discussion
 2017 Bell Museum Lake Pepin Freshwater Mussel Exhibit – Assisted design of display
 2016 Conservation Corps Youth Outdoors Career Day – Bell collection tours & career discussion

- 2016 Saturday with a Scientist – Public engagement of aquaponics at Bell Museum
 2015-26 50+ Collection tours – Bell Museum, Field Museum, & National Museum of Natural History

Online resources

- 2023- [MusselMapR](#) – R shiny web app for plotting US freshwater mussel distributions

School visits

- 2025 Natural history of freshwater mussels – School of Northern Lights
 2018 Cedar Creek Eco-Extravaganza – Jackson Middle School
 2016 Cedar Creek Eco-Extravaganza – Jackson Middle School

SERVICE

Service committees

- 2026 Central American Freshwater Mollusc Panel – IUCN Red List Review
 Antigua Guatemala: April 27-May 1 2026
 2025- Co-Chair of Genetics Committee – Freshwater Mollusk Conservation Society
 2025 Collections Manager of Insects Search Committee – Field Museum
 2024-25 Science & Scholarship Funding Committee – Field Museum
 2023 Invertebrate Zoology Collections Assistant Job Search – Field Museum
 2023-25 Invertebrate Zoology Collections Administrative Committee – Field Museum
 2023 Assistant Professor & Curator of Ichthyology Search Committee – University of Minnesota
 2023- Diversity, Equity, and Inclusion Committee – Freshwater Mollusk Conservation Society
 2018 Conservation Sciences Seminar Committee – University of Minnesota
 2017 Conservation Sciences Seminar Committee – University of Minnesota

Reviewer

Aquatic Conservation: Marine and Freshwater Ecosystems, Conservation Genetics, Diversity and Distributions, Ecological Applications, Ecology and Evolution, Ecosphere, Evolutionary Applications, Freshwater Mollusk Biology and Conservation, Freshwater Science, Hydrobiologia, Journal of Fish Biology, Journal of Heredity, Journal of Molluscan Studies, Journal of Zoological Systematics and Evolutionary Research, Malacologia, Scientific Reports, Zoological Journal of the Linnean Society, Zoomorphology

PROFESSIONAL DEVELOPMENT

- 2024 Leadership Training – Field Museum of Natural History
 2023 Strategic Planning for Biodiversity Collections – iDigBio
 2020 ECR²: Command-Line Bioinformatics – American Society of Naturalists Graduate Council
 2019 RADCamp (library prep, sequencing, & bioinformatics of 3RAD data) – Columbia University
 2012 PADI Search and Rescue SCUBA certification

FIELDWORK

Domestic

- 2012-26 Minnesota, Wisconsin
 2026 Virginia
 2023-25 Illinois
 2022 Maryland
 2018 Alabama, Missouri
 2017 Illinois, Iowa
 2016 Alabama, Arkansas, Illinois, Iowa, Mississippi, Tennessee

2014 Oregon

International

2026 Honduras – 15 day expedition (42 collection sites)

2024 Costa Rica – Co-led 14 day expedition (43 collection sites)

2023 Panama – 14 day expedition (38 collection sites)

2015 Canada

SOCIETY MEMBERSHIP

American Genetic Association

American Malacological Society

Freshwater Mollusk Conservation Society

Society for Integrative & Comparative Biology